

lecture 15.

Closures.

Reflexive.

R
 $R \cup \Delta$

$$\Delta = \{ (a,a) \mid a \in A \}.$$

P483.

$$R = \{ (1,1), (1,2), (2,1), (3,2) \}$$

$$A = \{ 1, 2, 3 \}.$$

$$\Delta = \{ (1,1), (2,2), (3,3) \}.$$

$$R \cup \Delta = \{ (1,1), (1,2), (2,1), (3,2), (2,2), (3,3) \}.$$

✓

Ex1:-

$$P483 \quad R = \{ (a,b) \mid a < b \}$$

$$A = \mathbb{Z}.$$

$$\Delta = \{ (a,a) \mid a \in A \}.$$

$$\Delta = \{ (-\infty, -\infty), (-\infty+1, -\infty+1), \dots, (-1, -1), (0,0), (1,1), \dots, (+\infty, +\infty) \}.$$

$$\Delta = \{ (a,b) \mid a \geq b \}.$$

$$R \cup \Delta = \{ (a,b) \mid a < b \vee a \geq b \}.$$

$$A = \{ (a,b) \mid a \leq b \}.$$

Symmetric

Closure.
 $R \cup R^{-1}$

$$R = \{ (1,1), (1,2), (2,1), (3,2) \}$$

$$A = \{ 1, 2, 3 \}.$$

$$R^{-1} = \{ (1,1), (2,1), (1,2), (2,3) \}.$$

$$R \cup R^{-1} = \{ (1,1), (1,2), (2,1), (3,2), (2,3) \}. \quad \checkmark$$

Ex2
483

$$R = \{ (a,b) \mid a > b \}.$$

$$A = \mathbb{Z}.$$

$$R^{-1} = \{ (b,a) \mid (a,b) \in R \}.$$

$$R^{-1} = \{ (b,a) \mid a > b \}$$

$$R^{-1} = \{(b, a) \mid \overline{a > b}\} \\ = \{(a, b) \mid a < b\}.$$

$$R \cup R^{-1} = \{(a, b) \mid a > b \vee a < b\} \\ u. = \{(a, b) \mid a \neq b\}.$$

Transitive Closure. (Not Straight forward).

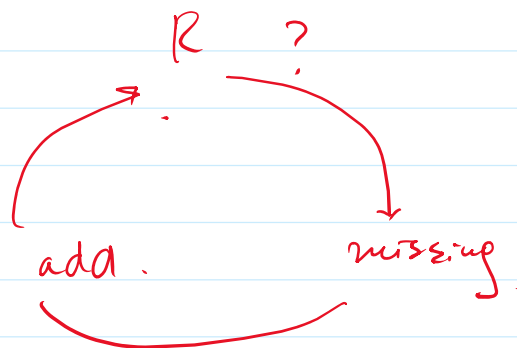
p484. $R = \{(1, 3), (1, 4), (2, 1), (3, 2)\}$.

$(1, 2), (2, 3), (2, 4), (3, 1)$. $\rightarrow$ missing.

$$R = \{(1, 3), (1, 4), (2, 1), (3, 2), (1, 2), (2, 3), (2, 4), (3, 1)\}.$$

$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$
 $b \quad c \quad a \quad b$

$(3, 4) \notin R.$

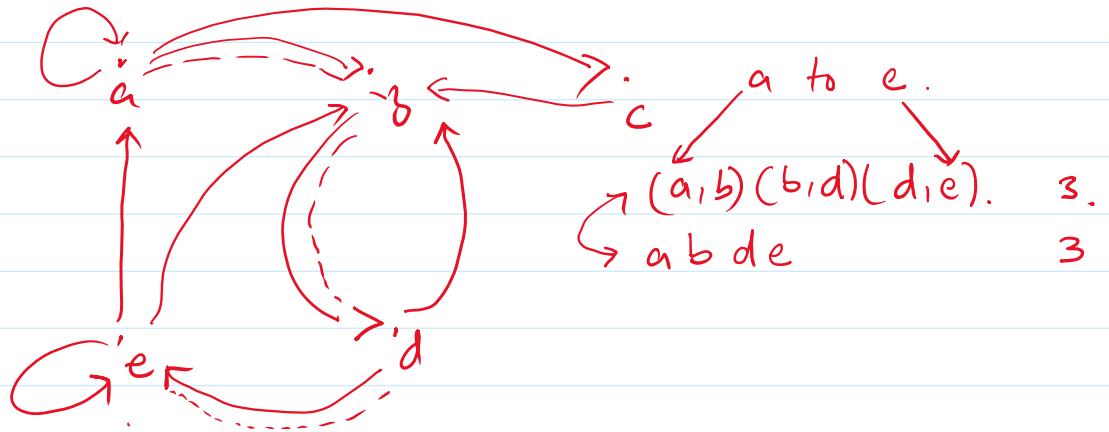


PATH: $\exists$ a path from "a" to "b" in a Graph. $\iff \exists$ a sequence of edges.

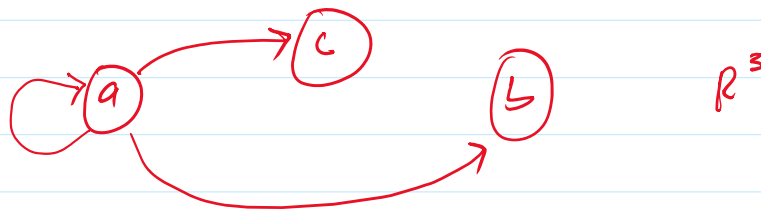
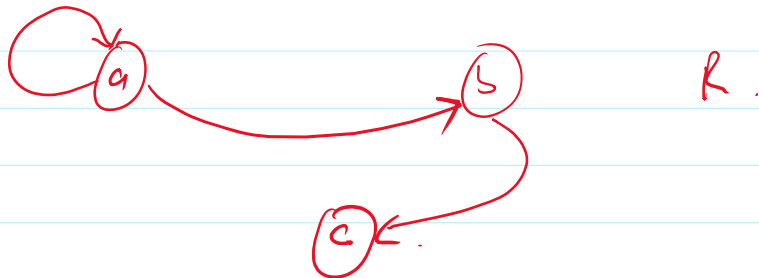
$$(a, x_1), (x_1, x_2), (x_2, x_3), \dots, (x_{n-1}, x_n), (x_n, b).$$

$(a, x_1), (x_1, x_2), (x_2, x_3), \dots, (x_{n-1}, x_n), (x_n, b).$

Ex 3
484

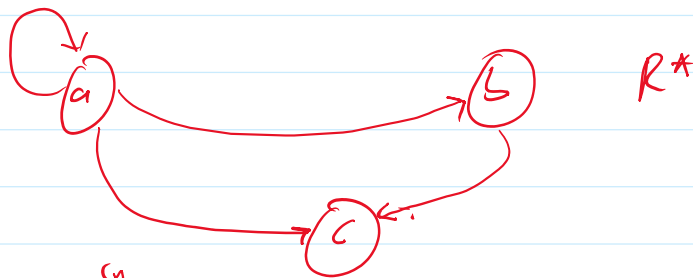
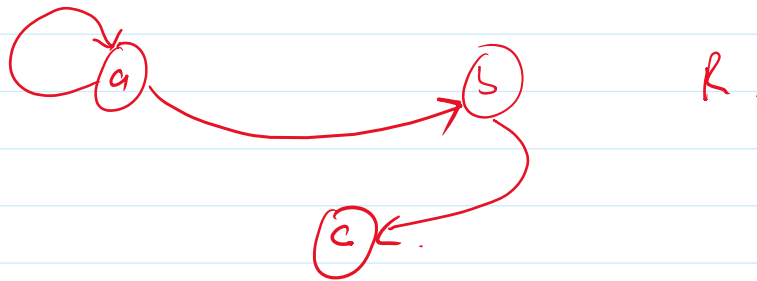


Theorem: Let R be a relation on A .
 485 $\exists$ a path of length $n \in \mathbb{Z}^+$
 from "a" to "b"
 if $(a,b) \in R^n$.



R^* Connectivity Relation.

R^* Connectivity Relation.
 $(a, b) \in R^*$ if $\exists$ a path of any length.



$$R^* = \bigcup_{i=1}^{\infty} R^i$$

$$R^*, R^1, R^{n-1}, \dots, R^1 \subseteq A \times A$$

$\mathbb{R} \times \mathbb{Y}$ $R = \{(a, b) \mid a \text{ has met } b\}$.

A : Set of people in world.

$R^2 = ?$

$R \circ R$.

So R .

$R^n = ?$

$(a, b) \in R$

$(x, y) \in R$

x_1 b

$A \times B$.

$B \times C$.

$R^* = ?$

$(a, c) \in R^*$

if $(a, x_1) \in R \wedge (x_1, c) \in R$

$(a, b) \in R^2$ a has met x_1 $\wedge$ x_1 has met b .

$$\begin{aligned} (a, b) &\in R^2 \\ (x_1, x_2) &\in R^2 \end{aligned}$$

$$\begin{aligned} A \times B \\ B \times C \end{aligned}$$

$(a, b) \in R^3$. a has met $x_1 \wedge x_1$ has met b .
 a has met $x_1 \wedge x_1$ has met $x_2 \wedge x_2$ has b .

R^n a has met $x_1 \wedge x_1$ has met x_2
 $\wedge \dots \wedge x_{n-1}$ has met b .

R^* $\exists$ any number of persons btw a & b
 who has met in sequence.

$R \times S$ $R \text{zd}(a, b) \mid a \text{ and } b \text{ has a}$
 common boundary.

Quiz 6

17-OCT-2015.

$R \text{zd}(a, b) \mid a - b \geq 0?$

Ans: 10, 105, 30?

a) find Graph of R .

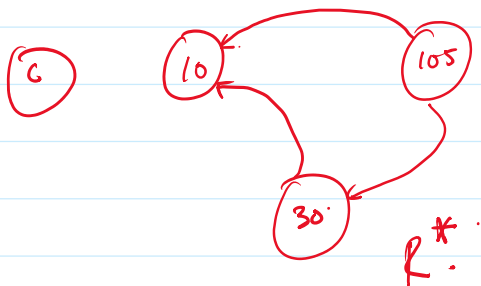
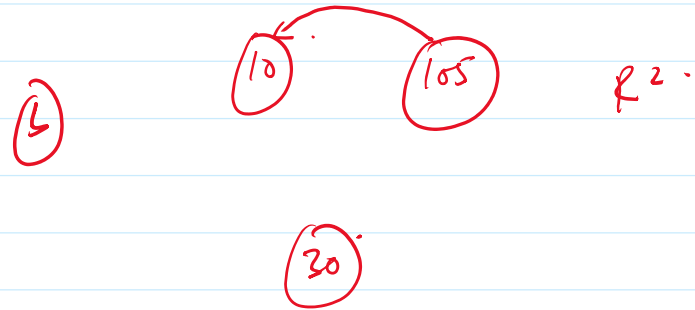
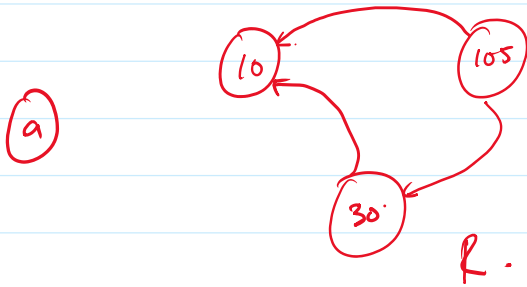
b) find R^2 of R in Graph form.

c) find R^* in matrix form.

d) find R in matrix form.

Solution -

Ans { 10, 105, 30 }



d

	10	105	30
10	0	0	0
105	1	0	1
30	1	0	0